

SMOHIT ARMAN SWAIN

192572092

CSA0802 PYTHON PROGRAMMING

ANALYICAL PROBLEM 1

Q1 Sum the given number upto single digit

5 marks

Write a Python code to find the sum of the given number upto single digit

input: 89789

output : 5

Your Code

```
1 num=int(input())
2 while num>=10:
3     s=0
4     while num>0:
5         s+=num%10
6         num//=10
7     num=s
8 print(num)
```

Input

89789

Output

5

1.

A numerical computation module is designed to generate a sequence where each number is the sum of the two preceding numbers. This sequence is known as the Fibonacci series.

Write a Python program that:

- Accepts a number n from the user
 - Generates and displays the first n terms of the Fibonacci sequence
 - Uses looping constructs to compute the sequence
- Ensure the program handles cases where n is less than or equal to zero.

Your Code

```
1 n=int(input())
2 if n<=0:
3     print("invalid input")
4 else:
5     a=0
6     b=1
7     for i in range(n):
8         print(a,end=" ")
9         c=a+b
10        a=b
11        b=c
```

Input

5

Output

0 1 1 2 3

2.

A text processing system receives user input in the form of a string that may contain alphabets, digits, and special characters.

Write a Python program that:

- Accepts a string input from the user
- Counts the number of vowels, consonants, digits, and special characters present in the string
- Displays the count for each category

Use appropriate string operations and control flow statements.

Your Code

```
1 s = input()
2
3 vowels=0
4 consonants=0
5 digits=0
6 special=0
7
8 for ch in s:
9     if ch.isalpha():
10         if ch.lower() in "aeiou":
11             vowels +=1
12         else:
13             consonants +=1
14     elif ch.isdigit():
```

3.

```
2
3 vowels=0
4 consonants=0
5 digits=0
6 special=0
7
8 for ch in s:
9     if ch.isalpha():
10         if ch.lower() in "aeiou":
11             vowels +=1
12         else:
13             consonants +=1
14     elif ch.isdigit():
15         digits +=1
16     else:
17         special+=1
18
19 print("Vowels=", vowels)
20 print("Consonants= ", consonants)
21 print("Digits=", digits)
22 print("Special Character= ", special)
```

Input

Hello@213

Output

```
Vowels= 2
Consonants= 3
Digits= 3
Special Character= 1
```

Output

```
Vowels= 2
Consonants= 3
Digits= 3
Special Character= 1
```

✓ Submitted

A program is designed to check whether a number entered by the user is even or odd. However, the program contains syntax and logical errors.

Analyze the given code and perform the following:

- Identify all errors present in the program
- Rewrite the corrected program
- Briefly justify each correction

```
num = input("Enter number: ")
```

```
if num % 2 == 0
```

```
print("Even")
```

```
else
```

```
print("Odd")
```

Your Code

```
1 num=int(input("enter number: "))
2 if num%2==0:
3     print("even")
4 else:
5     print("odd")
```

Input

5

Output

enter number: odd

4.

Print numbers from 1 to 10

A program is intended to display numbers from 1 to 10 using a loop. However, some parts of the code are missing.

Complete the program by filling in the blanks so that it correctly prints numbers from 1 to 10.

Program to print numbers from 1 to 10

for i in _____:

print(_____)

Also explain how the range() function works in this context.

Your Code

```
1 for i in range(1,11):
2     print(i)
```

Input

stdin input

Output

```
1
2
3
4
```

5.